

EEE4114F Class Test 1

March 2025

Duration: 1 hour

Total: 20 marks

Empl ID: _____

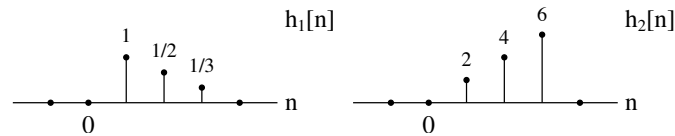
Instructions:

- This is a closed-book test.
- There are *four* questions totalling 20 marks. You must answer all of them.
- Questions should be answered in the space provided on the test script. If necessary, you may also write on the backs of the printed pages, but please indicate where you have done so.
- You must use a pen to complete the exam – pencil will only be marked for drawings. Values indicated on drawings must be written in pen.
- Please show all of your working clearly – your method and approach matter.
- Do not write your student number or name anywhere on this script; instead, provide only the 7-digit EMPLID found on your student card.
- A formula sheet is provided at the end of the paper; you may detach this, but please do so carefully so as not to damage the rest of the script.
- You have 1 hour.

Question	Marks	Grade
1	5	
2	5	
3	5	
4	5	
Total:	20	

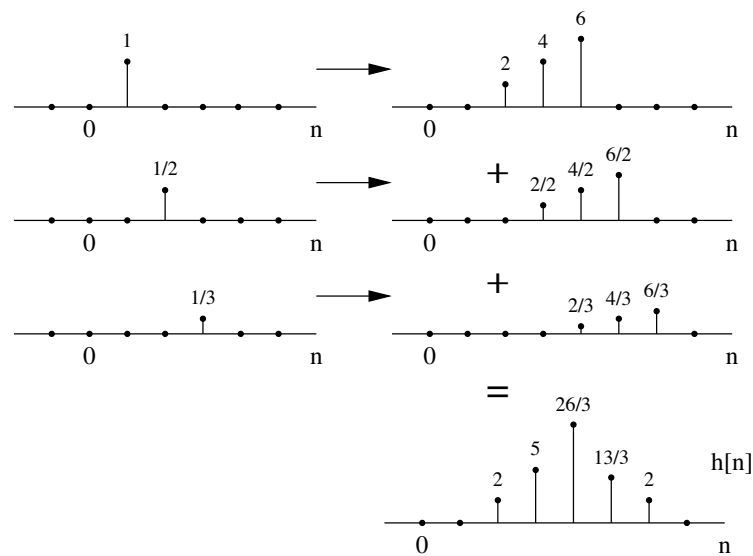
Question 1 [5 marks]

You are given a composite system that is formed by cascading two LTI processors with the following impulse responses:



Sketch the overall impulse response $h[n]$ for the composite system, and provide an analytical expression for it.

Solution: The overall impulse response of the composite system is the convolution of the two signals, which can easily be found by decomposing $h_1[n]$ into a linear sum of scaled and shifted impulses, computing the response from $h_2[n]$ for each impulse, and then adding the results. This is done as follows:



This can be written as an analytical expression as

$$h[n] = 2\delta[n-0] + 5\delta[n-1] + \frac{26}{3}\delta[n-2] + \frac{13}{3}\delta[n-3] + 2\delta[n-4]$$

Question 2 [5 marks]

Consider an LTI system represented by the difference equation

$$y[n] = x[n+1] - x[n]$$

where $x[n]$ and $y[n]$ are the input and output signals respectively.

1. Determine the impulse response $h[n]$ of the system, and thus determine whether the system is causal.

Solution: The impulse response $h[n]$ is the output when the input is $x[n] = \delta[n]$, so $h[n] = \delta[n+1] - \delta[n]$. And as $h[n]$ is not a purely right-sided signal, the system is not causal.

2. Find the step response of the system.

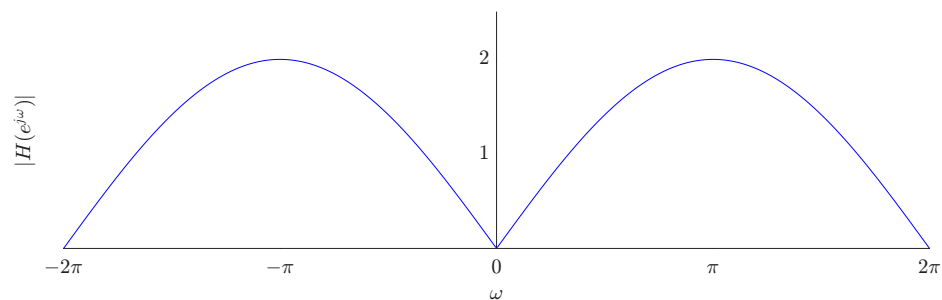
Solution: When the input is $x[n] = u[n]$ then the output will be $y[n] = u[n+1] - u[n] = \delta[n+1]$.

3. Find the frequency response $H(e^{j\omega})$ and plot its magnitude over the range $-2\pi \leq \omega \leq 2\pi$.

Solution: The frequency response is

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} (\delta[n+1] - \delta[n]) e^{-j\omega n} = e^{j\omega} - 1 = e^{j\omega/2} (e^{j\omega/2} - e^{-j\omega/2}) \\ &= 2je^{j\omega/2} \sin(\omega/2), \end{aligned}$$

so $|H(e^{j\omega})| = 2|\sin(\omega/2)|$. Plot as follows:



Question 3 [5 marks]

An LTI system function is provided below.

$$H(z) = \frac{8z - 4}{8z - 2}, \quad |z| > \frac{1}{4}.$$

Find the impulse response of an appropriate *causal* inverse system, and then comment on the stability of this inverse system.

Solution: The system function can be written as

$$H(z) = \frac{z - \frac{1}{2}}{z - \frac{1}{4}} = \frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{1}{4}z^{-1}}$$

The inverse system is then the reciprocal of $H(z)$, which we define as $G(z)$ such that $H(z)G(z) = 1$:

$$G(z) = \frac{z - \frac{1}{4}}{z - \frac{1}{2}} = \frac{1 - \frac{1}{4}z^{-1}}{1 - \frac{1}{2}z^{-1}} = \frac{1}{1 - \frac{1}{2}z^{-1}} - \frac{\frac{1}{4}z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

This has a pole at $z = \frac{1}{2}$, so the two possible ROCs are $|z| < \frac{1}{2}$ and $|z| > \frac{1}{2}$. Thus we choose $|z| > \frac{1}{2}$ so that the inverse system is causal (ROC extends out to infinity), which also causes the system to be stable (ROC contains the unit circle).

We can thus find the inverse system's impulse response by taking the inverse z-transform using tables and the time shift property:

$$g[n] = \left(\frac{1}{2}\right)^n u[n] - \frac{1}{4} \left(\frac{1}{2}\right)^{n-1} u[n-1].$$

Question 4 [5 marks]

You are given the system equation below with ROC $|z| > 0.5$:

$$H(z) = \frac{z^2}{z^2 + 0.25}$$

1. Find the system's impulse response.

Solution: We can exploit the following z-transform pair (with ROC $|z| > r$) from the tables:

$$r^n \cos(\omega_0 n) u[n] \quad \xleftrightarrow{\mathcal{Z}} \quad \frac{1 - r \cos(\omega_0) z^{-1}}{1 - 2r \cos(\omega_0) z^{-1} + r^2 z^{-2}}$$

By setting $\omega_0 = \pi/2$ and $r = 0.5$, we have:

$$(0.5)^n \cos((\pi/2)n) u[n] \quad \xleftrightarrow{\mathcal{Z}} \quad \frac{1 - 0.5 \cos(\pi/2) z^{-1}}{1 - 1 \cos(\pi/2) z^{-1} + (0.5)^2 z^{-2}}$$

$$(0.5)^n \cos((\pi/2)n) u[n] \quad \xleftrightarrow{\mathcal{Z}} \quad \frac{1}{1 + 0.25 z^{-2}}$$

$$(0.5)^n \cos((\pi/2)n) u[n] \quad \xleftrightarrow{\mathcal{Z}} \quad \frac{z^2}{z^2 + 0.25}$$

Since the right-hand side is $H(z)$, the left-hand side must be $h[n]$, and thus

$$h[n] = (0.5)^n \cos((\pi/2)n) u[n]$$

2. Sketch the system's pole-zero plot in the z-plane, and indicate the ROC.

Solution: The pole-zero plot should have two zeros at $z = 0$ (the origin), and poles at $z = \pm 0.5j$ (i.e., on the imaginary axis). As mentioned in the question, the ROC thus extends outwards from a circle (centred on the origin) of radius 0.5.